# Agnostic Learnability of Halfspaces via Logistic Loss

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## Setting.

**Binary classification:** given by distribution $P$ on $\mathbb{R}^d \times \{-1, +1\}$, we have i.i.d. samples from $P$.
**Linear classifier:** $x \mapsto \langle x, w \rangle$.
**Goal:** compete with the optimal solution $\bar{u}$, where

$$\mathcal{R}_{0-1}(\bar{u}) := \Pr_{(x,y)\sim P}\left(\mathrm{sign}(\langle \bar{u}, x\rangle) \neq y\right) = \mathrm{OPT} > 0.$$

**Logistic regression (LR):** let $\ell_{\log}(z) := \ln(1 + e^{-z})$, and let

$$\mathcal{R}_{\log}(w) := \mathbb{E}_{(x,y)\sim P}\left[\ell_{\log}\left(y\langle w, x\rangle\right)\right]$$

denote the population logistic risk of $w$ over $P$.

## Contributions.

Prior lower and upper bounds for LR do not match:

- General $P$: $1 - \mathrm{OPT}$ lower bound (Ben-David et al., 2012).
- Isotropic log-concave distributions: $\widetilde{\Omega}(\mathrm{OPT})$ lower bound (Diakonikolas et al., 2020).
- "Well-behaved" sub-exponential distributions: $\widetilde{O}(\sqrt{\mathrm{OPT}})$ upper bound with SGD (Frei et al., 2021).

**On "well-behaved" conditions:**

- Standard concentration and anti-concentration conditions.
- Example: a mixture of log-concave dist. (e.g., Gaussian).

Our lower and upper bounds for LR:

- $\Omega(\sqrt{\mathrm{OPT}})$ lower bound for well-behaved sub-exp dist.; matching $\widetilde{O}(\sqrt{\mathrm{OPT}})$ upper bound from (Frei et al., 2021).
- $\widetilde{O}(\mathrm{OPT})$ upper bound with "radial Lipschitzness."

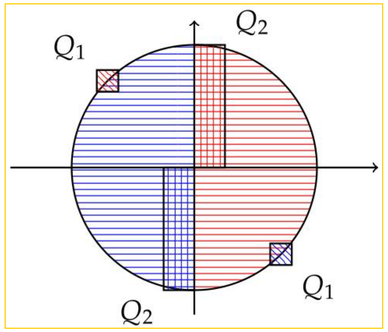
Upper bounds beyond LR:

- Diakonikolas et al. (2020) designed a nonconvex SGD that achieves $O(\mathrm{OPT}) + \epsilon$ risk using $\widetilde{O}(d/\epsilon^4)$ samples. They can also handle heavy-tailed distributions.
- Other prior algorithms achieving $O(\mathrm{OPT}) + \epsilon$ risk involve solving multiple rounds of convex program (Awasthi et al., 2014; Daniely, 2015).
- We design a simple two-phase convex program that achieves $O(\mathrm{OPT}\ln(1/\mathrm{OPT})) + \epsilon$ risk using $\widetilde{O}(d/\epsilon^2)$ samples, **without** radial Lipschitzness.

## Lower bound.

We construct a distribution $Q$ with three parts $Q_1, Q_2, Q_4$. (One part $Q_3$ is omitted for simplicity.)

1. $Q_1$ consists of two squares with edge length $\sqrt{\mathrm{OPT}/2}$ and density 1; one has center $(\sqrt{2}/2, -\sqrt{2}/2)$ and label $-1$, the other has center $(-\sqrt{2}/2, \sqrt{2}/2)$ and label $+1$.
2. $Q_2$ is supported on $[0, \sqrt{\mathrm{OPT}}] \times [0, 1]$ and $[-\sqrt{\mathrm{OPT}}, 0] \times [-1, 0]$ with density 1, and the label is given by $\mathrm{sign}(x_1)$.
3. $Q_4$ is the uniform distribution over the unit ball with density $q_4 := \frac{1-\mathrm{OPT}-2\sqrt{\mathrm{OPT}}}{\pi}$ and label $\mathrm{sign}(x_1)$.




**Theorem.** $Q$ is well-behaved and sub-exponential, and the global minimizer $w^*$ of $\mathcal{R}_{\log}$ has $\mathcal{R}_{0-1}(w^*) = \Omega(\sqrt{\mathrm{OPT}})$.

Proof ideas:

- To minimize $\mathcal{R}_{\log}$ on $Q_4$, we should go to infinity along $(1, 0)$.
- $Q_1$ keeps $w^*$ bounded with $\|w^*\| = O(1/\sqrt{\mathrm{OPT}})$.
- $Q_2$ drags $w^*$ above by an angle of $\Omega(\sqrt{\mathrm{OPT}})$.

## Upper bound: radial Lipschitzness.

To exclude $Q$ and overcome the lower bound, we make the following assumption for LR.

**Radial Lipschitzness:** There exists a measurable function $\kappa : \mathbb{R}_+ \to \mathbb{R}_+$ such that for any two-dimensional subspace $V$, letting $p_V$ denote the density of the projection of feature distribution onto $V$, then

$$|p_V(r, \theta) - p_V(r, \theta')| \leq \kappa(r)|\theta - \theta'|.$$

- Holds if $p_V$ is Lipschitz, even non-log-concave ones (e.g., Gaussian mixtures).
- Does not hold for general log-concave distributions.
- Does not hold for $Q$, non-Lipschitz near the vertical axis.

## Upper bound: logistic regression.

**Projected gradient descent (PGD):** let $w_0 := 0$, and

$$w_{t+1} := \Pi_{\mathcal{B}(1/\sqrt{\epsilon})}\left[w_t - \eta\nabla\widehat{\mathcal{R}}_{\log}(w_t)\right].$$

**Theorem.** If the dist. is well-behaved, sub-exponential and radially-Lipschitz, with learning rate $\widetilde{\Theta}(1/d)$, using $\mathrm{poly}(d, 1/\epsilon, \ln(1/\delta))$ samples and iterations, with prob. $1-\delta$, PGD outputs $w_t$ with

$$\mathcal{R}_{0-1}(w_t) = \widetilde{O}(\mathrm{OPT}) + \epsilon.$$

Proof ideas: under the given conditions, suppose $\hat{w}$ satisfies $\mathcal{R}_{\log}(\hat{w}) \leq \mathcal{R}_{\log}(\|\hat{w}\|\bar{u}) + \epsilon'$, then

$$\mathcal{R}_{0-1}(\hat{w}) = \widetilde{O}\left(\max\left\{\mathrm{OPT}, \sqrt{\frac{\epsilon'}{\|\hat{w}\|}}, \frac{C_\kappa}{\|\hat{w}\|^2}\right\}\right). \tag{1}$$

- $C_\kappa = O(\ln(1/\mathrm{OPT})^2)$ for Lipschitz density.
- The global optimizer $w^*$ of $\mathcal{R}_{\log}$ satisfies $\|w^*\| = \widetilde{\Omega}(1/\sqrt{\mathrm{OPT}})$; PGD also ensures $w_t$ has small $\epsilon'$ and $\|w_t\| = \widetilde{\Omega}(1/\sqrt{\mathrm{OPT}})$.

## Upper bound: two-phase convex program.

Motivation: (1) holds for the hinge loss $\ell_h(z) := \max\{-z, 0\}$ **without** radial Lipschitzness!

$$\mathcal{R}_{0-1}(\hat{w}) = \widetilde{O}\left(\max\left\{\mathrm{OPT}, \sqrt{\frac{\epsilon'}{\|\hat{w}\|}}\right\}\right).$$

However, the global minimizer of $\mathcal{R}_h$ is 0; if we find $\hat{w}$ by global minimization, as $\epsilon'$ gets small, $\|\hat{w}\|$ also gets small.

Ideas: LR followed by restricted Perceptron:

- LR finds unit $v$ that is $\widetilde{O}(\sqrt{\mathrm{OPT}})$ away from $\bar{u}$.
- Then minimize $\mathcal{R}_h$ over $\mathcal{D} := \{w | \langle w, v\rangle \geq 1\}$.
  - $\forall w \in \mathcal{D}, \|w\| \geq 1$.
  - $\|\hat{w}\|\bar{u}$ may not in $\mathcal{D}$, but $\left(1 + \widetilde{O}(\mathrm{OPT})\right)\|\hat{w}\|\bar{u} \in \mathcal{D}$!
- Use SGD for better sample complexity.

**Theorem.** If the dist. is well-behaved and sub-exponential, using $\widetilde{O}(d/\epsilon^2)$ samples, SGD can achieve zero-one risk $O(\mathrm{OPT}\ln(1/\mathrm{OPT})) + \epsilon$.